

5. CO_2

6. (i) Na_2CO_3 , CO_2

7. (3) $B > A > C$ - decreasing acidic strength

8. (a) A is false but R is true

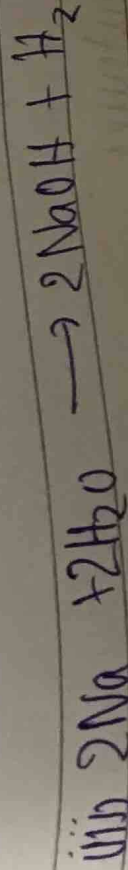
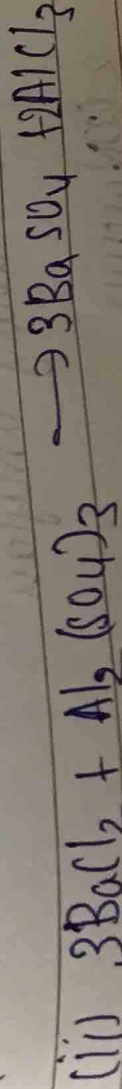
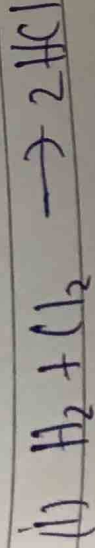
9. Alkali substances are those which can be dissolved in an aqueous medium whereas bases cannot be dissolved.

10. HCl is oxidised to Cl_2

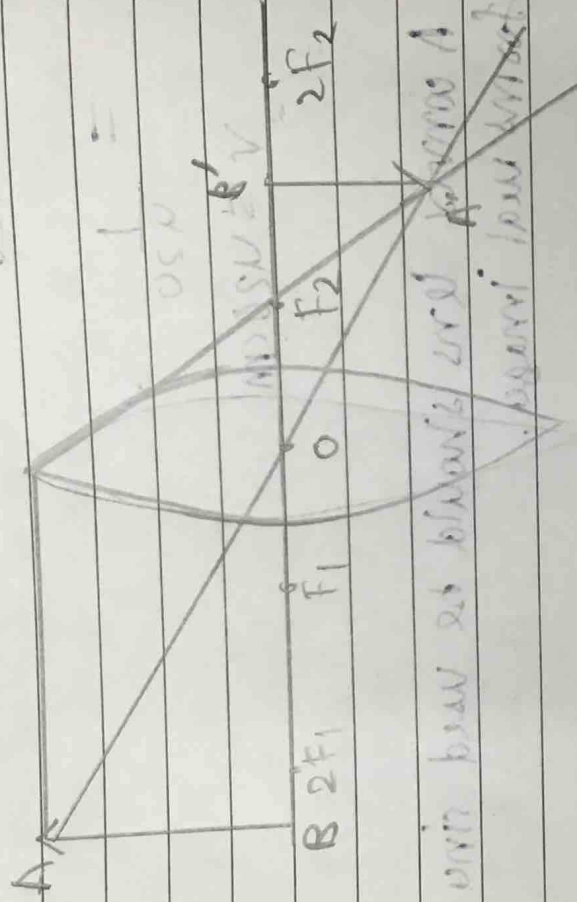
MnO_2 is oxidising agent

MnO_2 is reduced to MnCl_2

HCl is reducing agent



For diminished image the candle should be placed at a distance greater than $2F$ from lens.



Chemistry

1. (2) PbS - Reductant; H_2O_2 - Oxidant
2. (1) CaCO_3 , Ca(OH)_2
3. (1) $\text{X} - \text{NaOH}$ $\gamma - \text{CO}_2$
4. (1) Na_2CO_3 , CO_2 , tartaric acid, NaHCO_3

12. X is CaCO_3 - Calcium carbonate

Y is Ca(OH)_2 - slaked lime

Z is CaOCl_2 - Bleaching powder

G is Cl_2 - Chlorine gas

13.

(i) The solution will turn light green from blue due to formation of ferrous sulphate

(ii) Iron nail will develop a reddish brown layer of

copper metal

(iii) White

(iv) Dil. HCl decomposes BaCl_2 , SO_2 and H_2O . Since BaCl_2 is soluble in water. White precipitate disappears.

Biology

1. (2) Salivary amylase

2. (3) Nostoc

3. (1) Antennules